

TKR COLLEGE OF ENGINEERING & TECHNOLOGY

SPARK EXPO-2026

Project Report

Submitted by

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B.Tech CSD

UNDER THE GUIDANCE OF

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TKR COLLEGE OF ENGINEERING AND TECHNOLOGY
AUTONOMOUS

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CERTIFICATE

This is to certify that **MADDI DURGA PRASAD** bearing Roll No. **25K95A6709** is a bonafide student of **TKR COLLEGE OF ENGINEERING & TECHNOLOGY** submitted the project report on “**E-COMMERCE SALES & CUSTOMER ANALYSIS USING MYSQL AND POWER BI**” in partial fulfillment of the requirement for the award of **Bachelor of Technology** degree in **Computer Science & Engineering (DATA SCIENCE)** of the **Jawaharlal Nehru Technological University, Hyderabad**, during the academic year 2025-2026, is a bonafide record of work carried out under our guidance and supervision. The results presented in this dissertation have been verified and are found to be satisfactory. The results embodied in this dissertation have not submitted by any other university for the award of any other degree or diploma.

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Class, Incharge

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HOD Name

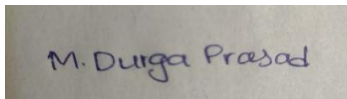
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TKR COLLEGE OF ENGINEERING & TECHNOLOGY

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DECLARATION

I hereby declare that the project entitled "**E-COMMERCE SALES & CUSTOMER ANALYSIS USING MYSQL AND POWER BI**". submitted for the B. Tech. (CSD) Degree is my original work and the project has not formed the basis for the award of any degree, associate ship, fellowship or any similar titles.

A rectangular box containing a handwritten signature in blue ink that reads "M. Durga Prasad".

Signature of the Student

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Place: HYDERABAD

ACKNOWLEDGEMENT

I would like to take this opportunity to express my heartfelt gratitude and my sincere thanks to **Prof.Tejashwini** , **Class Incharge of CSE(DATA SCIENCE) Section-B**, for her valuable advice and guidance for developing the project & helped me for the successful completion of my project on " **E-COMMERCE SALES & CUSTOMER ANALYSIS USING MYSQL AND POWER BI** ".

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MADDI DURGA PRASAD

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**E-COMMERCE SALES & CUSTOMER ANALYSIS USING
MYSQL AND POWER BI**

ABSTRACT

This project presents a comprehensive E-Commerce Sales and Customer Analysis solution leveraging MySQL for robust data management and Power BI for interactive business intelligence and visualization. The primary objective was to transform raw transactional data into actionable insights to optimize sales performance, enhance customer segmentation, and improve overall business decision-making in a competitive e-commerce environment. Data from multiple sources—including orders, customers, products, and returns—was ingested and stored in a well-normalized MySQL relational database. Complex SQL queries were developed to clean the data, handle missing values, calculate key performance indicators (KPIs) such as Revenue, Average Order Value (AOV), Customer Lifetime Value (CLV), and Retention Rate, and perform advanced analytics including RFM (Recency, Frequency, Monetary) segmentation and cohort analysis.

Keywords: E-Commerce Analytics, Sales Analysis, Customer Segmentation, MySQL, Power BI, RFM Analysis, Business Intelligence.

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CHAPTER-1

INTRODUCTION

The emergence of e-commerce has revolutionized the way businesses operate and interact with customers in the modern digital economy. With the rapid growth of online shopping platforms, enormous amounts of data are being generated daily through customer transactions, website interactions, and purchasing behaviors. This massive volume of data holds immense potential for deriving valuable insights that can drive business growth and competitiveness. However, many e-commerce companies still struggle to effectively analyze and utilize this data for strategic decision-making.

In today's highly competitive marketplace, understanding sales patterns, customer preferences, and market trends has become crucial for survival and sustainable growth. Traditional reporting methods are no longer sufficient to handle the complexity and velocity of e-commerce data. There is a growing need for advanced data analytics solutions that can transform raw data into meaningful and actionable business intelligence.

This project, titled **“E-Commerce Sales and Customer Analysis using MySQL and Power BI”**, aims to address these challenges by developing a comprehensive analytics system. The project focuses on efficient data storage and management using MySQL relational database and powerful data visualization using Microsoft Power BI.

The primary objective of this project is to analyze historical sales data to uncover hidden patterns, identify key performance indicators, segment customers effectively, and provide interactive dashboards for better decision-making. By leveraging the strong querying capabilities of MySQL and the rich visualization features of Power BI, this system will help stakeholders gain deep insights into sales performance, customer lifetime value, product popularity, and regional trends.

The scope of the project includes data cleaning, database design, implementation of complex SQL queries, RFM-based customer segmentation, and the creation of interactive dashboards. This end-to-end solution bridges the gap between data storage and business intelligence, making it accessible even for users with limited technical expertise.

In summary, this project demonstrates the practical application of database management and business intelligence tools in solving real-world e-commerce problems. It showcases how proper data analysis can lead to improved marketing strategies, better inventory management, enhanced customer satisfaction, and increased revenue.

The successful implementation of this project will not only fulfill academic requirements but also provide a ready-to-use framework that can be adopted by small and medium e-commerce businesses looking to harness the power of data analytics.

1.2 SCOPE OF THE PROJECT

The **E-Commerce Sales & Customer Analysis** project aims to analyze online retail data using **MySQL** for data storage and querying, and **Power BI** for interactive visualization and reporting. The project helps businesses understand sales performance, customer behavior, product trends, and regional performance through data-driven insights.

Store and manage e-commerce data efficiently using **MySQL**.

Analyze sales performance across different products, categories, and regions.

Identify top-selling and low-performing products.

Study customer purchasing behavior and buying patterns.

Track revenue, profit, quantity sold, and order trends over time.

Monitor regional and state-wise sales performance.

Create interactive dashboards using **Power BI** for real-time business insights.

Support data-driven decision-making for improving sales and customer satisfaction.

Provide business intelligence reports for management and stakeholders.

Build a scalable analytics solution that can be enhanced with predictive analytics and machine learning in the future.

1.3 OBJECTIVE

The main objective of this project is to analyze e-commerce sales and customer data using **MySQL** and **Power BI** to generate meaningful business insights. It aims to help organizations understand sales trends, customer behavior, and product performance through interactive dashboards and reports.

Store and manage e-commerce data using **MySQL**.

Analyze sales, revenue, profit, and order performance.

Identify top-performing and low-performing products.

Understand customer purchasing behavior and preferences.

Evaluate sales performance across different regions and categories.

Create interactive dashboards for easy data visualization.

Support data-driven decision-making for business growth.

Improve business performance through actionable insights.

Generate accurate reports for management and stakeholders.

Enhance analytical skills using **SQL** and **Power BI** in a real-world business scenario.

1.4 EXISTING SYSTEM

In the existing system, e-commerce sales data is often stored in multiple files or databases and analyzed manually using spreadsheets or basic reporting tools. Business users generate reports separately, making it difficult to obtain a complete view of sales performance, customer behavior, and product trends. Decision-making is based on static reports, which are time-consuming to prepare and update.

1.4.1 EXISTING SYSTEM DISADVANTAGES

- Manual data analysis is time-consuming and error-prone.
- Data is scattered across multiple sources, making analysis difficult.

- Lack of real-time insights into sales and customer performance.

1.5 PROPOSED SYSTEM

The proposed system uses **MySQL** for efficient data storage and management, and **Power BI** for interactive data visualization and business intelligence. It integrates sales and customer data into a centralized database, enabling users to analyze sales performance, customer behavior, product trends, and regional performance through dynamic dashboards. The system provides accurate, real-time insights that support faster and more effective business decision-making.

1.6.1 PROPOSED SYSTEM ADVANTAGES

- Centralized storage and management of e-commerce data.
- Fast and efficient data analysis using SQL queries.
- Interactive dashboards for real-time business insights.
- Scalable system that can be enhanced with predictive analytics and AI in the future.
- Supports informed and data-driven decision-making.

CHAPTER 2

PROJECT DESCRIPTION

2.1 GENERAL:

The **E-Commerce Sales & Customer Analysis Using MySQL and Power BI** project is a Business Intelligence (BI) solution designed to analyze and visualize e-commerce sales data. The project focuses on transforming raw sales and customer information into meaningful insights that help organizations improve business performance and make data-driven decisions.

The system uses **MySQL** as the backend database to store, manage, and query e-commerce data efficiently. SQL queries are used to extract, filter, and analyze information such as sales, revenue, profit, customer details, product categories, and regional performance. The processed data is then connected to **Power BI**, where interactive dashboards and visual reports are created.

The dashboard provides a comprehensive view of key business metrics, including total sales, total profit, total orders, customer distribution, product performance, category-wise sales, regional analysis, and monthly sales trends. Users can apply filters and slicers to explore the data from different perspectives and identify important business patterns.

2.2 METHODOLOGIES

2.2.1 MODULES NAME:

- **Data Collection**
- **Data Storage**
- **Data Cleaning**
- **Data Analysis Using SQL**
- **Data Integration with Power BI**
- **Dashboard Development**
- **Report Generation**
- **Business Decision Support**

2.2.2 MODULES EXPLANATION:

1. Data Collection

- Collect e-commerce sales data containing customer details, product information, orders, revenue, profit, quantity, and regional data.

2. Data Storage

- Import and organize the collected data into a MySQL database using well-structured tables and relationships.

3. Data Cleaning

- Remove duplicate records, handle missing values, correct inconsistencies, and ensure data accuracy before analysis.

4. Data Analysis Using SQL

- Write SQL queries to calculate sales, profit, revenue, customer statistics, product performance, category-wise analysis, and regional sales trends.

5. Data Integration with Power BI

- Connect the MySQL database to Power BI to import analyzed data for visualization.

6. Dashboard Development

- Design interactive dashboards with charts, KPIs, tables, maps, slicers, and filters for easy exploration of business data.

7. Report Generation

- Generate dynamic reports that provide insights into sales performance, customer behavior, product trends, and regional analysis.

8. Business Decision Support

- Use the generated insights to support strategic planning, improve sales performance, optimize inventory, and enhance customer satisfaction.

2.3 TECHNIQUE USED OR ALGORITHM USED

2.3.1 EXISTING TECHNIQUE: -

The existing technique for e-commerce sales analysis mainly relies on **manual reporting methods**, spreadsheets, and basic database queries. Organizations typically store sales data in multiple files or simple databases and generate reports manually using tools such as Microsoft Excel. This approach provides only limited insights and requires significant time and effort to analyze business performance.

The analysis is generally based on static reports without interactive visualization, making it difficult to identify trends, customer behavior, and product performance. As the volume of e-commerce data increases, these traditional techniques become inefficient and less effective for decision-making.

2.3.2 PROPOSED TECHNIQUE USED OR ALGORITHM USED:

The proposed system uses a combination of **MySQL** for data storage and analysis and **Power BI** for data visualization and business intelligence. Instead of relying on manual reporting, the system follows a structured data analytics approach that enables efficient processing, analysis, and presentation of e-commerce sales data.

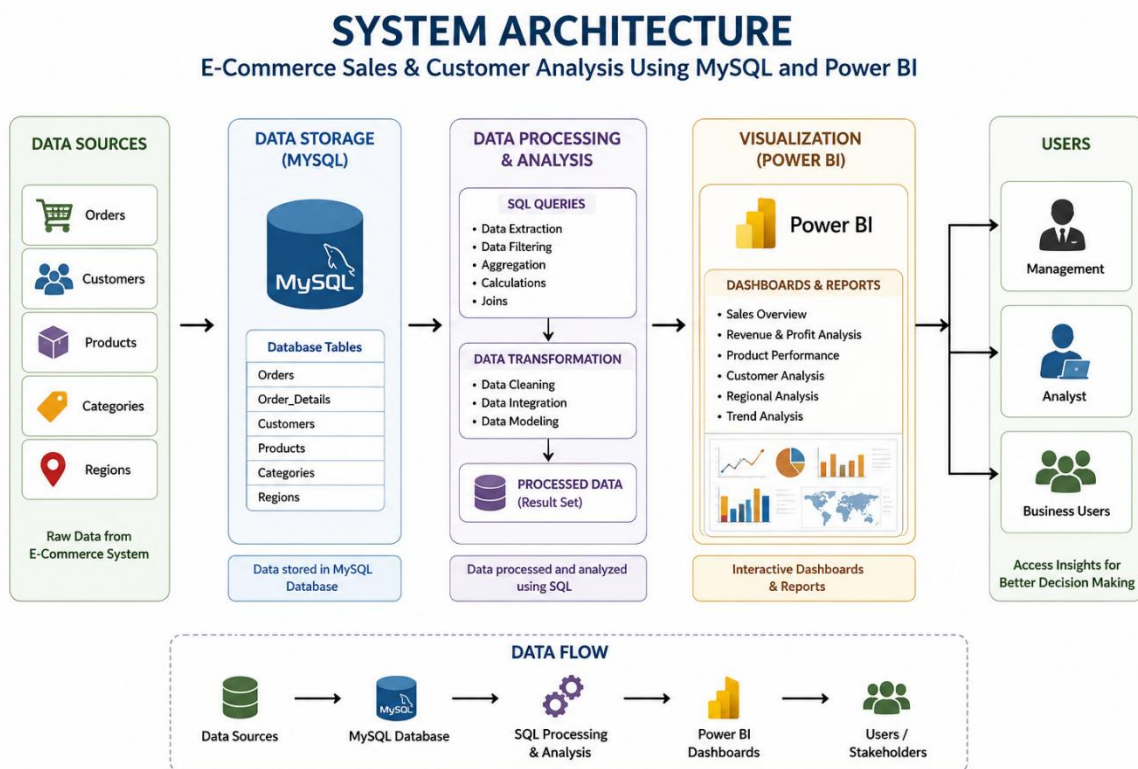
SQL queries are used to retrieve, filter, aggregate, and analyze data from the MySQL database. The processed data is then connected to Power BI, where interactive dashboards, charts, KPIs, and reports are created. This technique allows users to monitor business performance, identify sales trends, analyze customer behavior, and make informed decisions through dynamic visualizations.

Techniques Used

- MySQL Database Management for storing and managing e-commerce data.
- SQL Queries for data extraction, filtering, aggregation, and analysis.

- Data Cleaning and Transformation to improve data quality and consistency.
- Power BI Data Modeling to establish relationships between tables.
- Interactive Dashboarding using charts, cards, slicers, maps, and KPIs.
- Business Intelligence (BI) techniques for generating actionable insights.

2.4 SYSTEM ARCHITECTURE



CHAPTER 3

REQUIREMENTS ENGINEERING

3.1 GENERAL

Requirements Engineering is the process of identifying, analyzing, documenting, and validating the functional and non-functional requirements of a software system. It ensures that the developed system meets the needs of users and supports business objectives effectively.

In the E-Commerce Sales & Customer Analysis Using MySQL and Power BI project, the primary requirement is to collect, store, analyze, and visualize e-commerce sales data to generate meaningful business insights. The system integrates MySQL for efficient database management and Power BI for interactive reporting and dashboard creation.

3.2 HARDWARE REQUIREMENTS

The hardware requirements serve as the foundation for implementing the proposed system. These specifications ensure smooth execution of document processing, AI interaction, and learning resource generation.

- PROCESSOR : Intel Core i5 (8th Gen)
- RAM : 8GB RAM
- HARD DISK : 256 GB

3.3 SOFTWARE REQUIREMENTS

The following software components are required to develop and execute the E-Commerce Sales & Customer Analysis Using MySQL and Power BI project..

Software	Purpose
Operating System	Windows 10/11 (64-bit)
Database	MySQL Server 8.0 or above
Database Client	MySQL Workbench 8.0
Business Intelligence Tool	Microsoft Power BI Desktop
Programming/Query Language	SQL
Spreadsheet Tool	Microsoft Excel (for importing/exporting datasets)
Data Source Format	CSV / Excel (.xlsx)
Documentation Tool	Microsoft Word
Presentation Tool	Microsoft PowerPoint
Version Control (Optional)	Git & GitHub

3.4 FUNCTIONAL REQUIREMENTS

Functional requirements describe the specific features and operations that the E-Commerce Sales & Customer Analysis Using MySQL and Power BI system must perform to meet user and business needs.

Functional Requirements

1. Data Import

- The system should import e-commerce data from CSV or Excel files into the MySQL database.

2. Data Storage

- The system should store customer, product, order, sales, and regional data in a structured MySQL database.

3. Data Management

- The system should allow efficient insertion, updating, retrieval, and management of data using SQL queries.

4. Sales Analysis

- The system should analyze sales, revenue, profit, quantity sold, and order performance.

5. Customer Analysis

- The system should analyze customer purchasing behavior, order frequency, and customer distribution.

6. Product Analysis

- The system should identify top-selling products, low-performing products, and category-wise sales performance.

7. Regional Analysis

- The system should generate state-wise and region-wise sales and profit reports.

8. Dashboard Generation

- The system should create interactive dashboards using Power BI with charts, KPIs, maps, slicers, and filters.

9. Report Generation

- The system should generate dynamic reports for business analysis and decision-making.

10. Data Visualization

- The system should display business insights using graphs, bar charts, line charts, pie charts, and tables.

11. Search and Filter

- The system should allow users to filter and analyze data based on region, category, customer, product, and time period.

12. Performance Monitoring

- The system should monitor key performance indicators (KPIs) such as Total Sales, Total Profit, Total Orders, and Average Order Value.

3.5 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality attributes and performance standards of the E-Commerce Sales & Customer Analysis Using MySQL and Power BI system. These requirements ensure that the system is efficient, reliable, secure, and easy to use.

Non-Functional Requirements

1. Performance

- The system should retrieve and analyze data efficiently.
- Dashboards should load quickly and display results with minimal delay.
- SQL queries should execute within an acceptable response time.

2. Reliability

- The system should provide accurate and consistent analytical results.
- Data should be stored securely without corruption or loss.
- The system should operate continuously with minimal downtime.

3. Scalability

- The system should support increasing volumes of e-commerce data.
- Additional datasets and dashboards can be added without affecting performance.

4. Security

- Access to the database should be restricted to authorized users.
- Sensitive business data should be protected from unauthorized access.
- Regular database backups should be maintained.

5. Usability

- The Power BI dashboard should have a simple and user-friendly interface.
- Reports should be easy to understand and navigate.
- Interactive filters and slicers should improve the user experience.

6. Maintainability

- The database structure and SQL queries should be well organized.
- The system should be easy to update, modify, and maintain.
- New reports and visualizations can be added with minimal effort.

7. Availability

- The system should be available whenever business users need to access reports.
- Dashboards should provide consistent access to business insights.

8. Accuracy

- The system should generate precise calculations for sales, revenue, profit, and customer analysis.
- Reports should reflect accurate and up-to-date data.

9. Compatibility

- The system should work seamlessly with MySQL, Power BI, and Microsoft Excel.
- It should support standard CSV and Excel data formats.

CHAPTER 4

DEVELOPMENT TOOLS

4.1 Development Tools

The E-Commerce Sales & Customer Analysis Using MySQL and Power BI project was developed using modern database management and business intelligence tools. These tools help in storing, managing, analyzing, and visualizing e-commerce data efficiently.

1. MySQL

- Purpose: Database Management System (DBMS)
- Description: MySQL is an open-source relational database used to store, organize, and manage e-commerce sales and customer data. It supports SQL queries for data retrieval, filtering, aggregation, and analysis.

2. MySQL Workbench

- Purpose: Database Development Tool
- Description: MySQL Workbench is used to create databases, design tables, write and execute SQL queries, manage database relationships, and perform administrative tasks.

3. Microsoft Power BI

- Purpose: Business Intelligence and Data Visualization
- Description: Power BI is used to connect with the MySQL database and create interactive dashboards, charts, KPIs, maps, and reports. It helps convert raw data into meaningful business insights.

4. Microsoft Excel

- Purpose: Data Preparation
- Description: Excel is used to import, clean, and organize datasets before loading them into the MySQL database. It also supports CSV and Excel file formats for data exchange.

5. SQL (Structured Query Language)

- Purpose: Data Query and Analysis
- Description: SQL is used to create database tables, insert records, retrieve data, perform joins, calculate business metrics, and generate analytical reports.

6. Microsoft Word

- Purpose: Documentation
- Description: Microsoft Word is used to prepare the project report, documentation, and technical descriptions.

7. Microsoft PowerPoint

- Purpose: Presentation
- Description: PowerPoint is used to create project presentations, summarize findings, and demonstrate dashboards during project reviews.

4.2 History of MySQL

MySQL is one of the world's most popular open-source Relational Database Management Systems (RDBMS). It was developed in 1995 by Michael Widenius, David Axmark, and Allan Larsson at the Swedish company MySQL

AB. The name "MySQL" is derived from the combination of "My", the name of Michael Widenius's daughter, and "SQL" (Structured Query Language).

MySQL was designed to provide a fast, reliable, and easy-to-use database solution for web applications and enterprise systems. Due to its high performance, scalability, and open-source nature, it quickly became a preferred database for websites, e-commerce platforms, and business applications.

In 2008, MySQL AB was acquired by Sun Microsystems, and later in 2010, Oracle Corporation acquired Sun Microsystems, making Oracle the owner and maintainer of MySQL. Today, MySQL continues to be one of the most widely used database systems in industries, educational institutions, and software development.

4.3 Importance of MySQL

MySQL plays a vital role in managing and analyzing business data efficiently. It provides a secure and reliable platform for storing large volumes of structured information and supports powerful SQL queries for data retrieval and analysis.

Importance of MySQL

- Open-source and freely available.
- Efficient storage and management of large datasets.
- Fast execution of SQL queries.
- High reliability and data integrity.
- Supports multiple users simultaneously.
- Provides strong security features and access control.

- Easy integration with Business Intelligence tools like Power BI.
- Scalable for small, medium, and large applications.
- Supports backup and recovery of databases.
- Widely used in web applications, e-commerce, and enterprise systems.

4.4 Features of MySQL

MySQL offers several advanced features that make it one of the most popular relational database management systems.

Features of MySQL

- Open-source Relational Database Management System (RDBMS).
- Uses Structured Query Language (SQL).
- High-speed data processing and query execution.
- Supports large databases with millions of records.
- Multi-user and multi-threaded architecture.
- Strong security with user authentication and privileges.
- Supports transactions and ACID properties.
- Backup and recovery support.
- Cross-platform compatibility (Windows, Linux, macOS).
- Easy integration with Power BI, Excel, Python, Java, PHP, and other applications.
- Data indexing for faster retrieval.

- Supports views, stored procedures, triggers, and functions.
- Reliable performance with minimal maintenance.

4.5 Importance of Power BI

Power BI is a Business Intelligence (BI) and data visualization tool developed by Microsoft. It helps organizations convert raw data into meaningful insights through interactive dashboards and reports. Power BI enables businesses to monitor performance, identify trends, and make informed decisions based on real-time data analysis.

Importance of Power BI

- Converts complex data into interactive visual dashboards.
- Supports real-time business monitoring and reporting.
- Connects with multiple data sources, including MySQL, Excel, and SQL Server.
- Provides interactive charts, graphs, maps, and KPIs.
- Enables faster and more accurate business decision-making.
- Simplifies data analysis through drag-and-drop features.
- Supports data filtering, slicing, and drill-down analysis.
- Helps identify sales trends, customer behavior, and business opportunities.
- Improves reporting efficiency by reducing manual work.
- Widely used in Business Intelligence, Data Analytics, and Enterprise Reporting.

CHAPTER 5

IMPLEMENTATION

5.1 GENERAL

Coding:

```
CREATE DATABASE ecommerce_db;
```

```
USE ecommerce_db;
```

```
CREATE TABLE customers(customer_id INT PRIMARY KEY, name  
VARCHAR(100),city VARCHAR(50));
```

```
INSERT INTO customers VALUES(0001,'RAVI','HYD');
```

```
INSERT INTO customers VALUES(0002,'RIYA','HYD');
```

```
INSERT INTO customers VALUES(0003,'SURYA','HYD');
```

```
INSERT INTO customers VALUES(0004,'LAASYA','CHENNAI');
```

```
INSERT INTO customers VALUES(0005,'SANDHYA','HYD');
```

```
INSERT INTO customers VALUES(0006,'RAVI.K','HYD');
```

```
INSERT INTO customers VALUES(0007,'NAVI','HYD');
```

```
INSERT INTO customers VALUES(0008,'M.RAVI','DELHI');
```

```
INSERT INTO customers VALUES(0009,'DEVA','MUMBAI');
```

```
INSERT INTO customers VALUES(0010,'SONNU','MUMBAI');
```

```
INSERT INTO customers VALUES(0011,'SWAPNA','DELHI');

INSERT INTO customers VALUES(0012,'SURESH','AP');

INSERT INTO customers VALUES(0013,'E.DURGA','HYD');

INSERT INTO customers VALUES(0014,'KUMAR','AP');

INSERT INTO customers VALUES(0015,'K.RAVI','DELHI');

INSERT INTO customers VALUES(0016,'RISHI','DELHI');

INSERT INTO customers VALUES(0017,'SWETHA','DELHI');

INSERT INTO customers VALUES(0018,'GOWTHAM','MUMBAI');

INSERT INTO customers VALUES(0019,'BUNNY','HYD');

INSERT INTO customers VALUES(0020,'JAY','DELHI');
```

.....

```
SELECT customer_id FROM customers
```

```
WHERE city = 'HYD';
```

```
SELECT c.city,
```

```
SUM(o.total_amount) AS total_revenue
```

```
FROM orders o
```

```
JOIN customers c ON o.customer_id = c.customer_id
```

```
WHERE c.city = 'Patna'
```

```
ORDER BY total_revenue DESC;
```

```
UPDATE customers
```

```
SET city = 'Madhya Pradesh'
```

```
WHERE customer_id IN (0019,0013,0007,0006,0005,0003,0002,0001,0022);
```

```
UPDATE customers
```

```
SET city = 'BIHAR'
```

```
WHERE customer_id IN (0014,0012);
```

```
SELECT*FROM customers;
```

```
INSERT INTO customers VALUES(0021,'RASHITHA','DELHI');
```

```
INSERT INTO customers VALUES(0023,'VAIBHAV','PUNE');
```

```
INSERT INTO customers VALUES(0024,'ASISNETHA','PUNE');
```

```
INSERT INTO customers VALUES(0025,'VAMSHI','ASSAM');
```

```
INSERT INTO customers VALUES(0026,'RUPA','DELHI');
```

```
.....
```

```
SELECT * FROM customers;
```

```
CREATE TABLE products(product_id INT PRIMARY KEY,product_name  
VARCHAR(100),price DECIMAL(10,2));
```

```
INSERT INTO products (product_id, product_name, price) VALUES
```

(101, 'Laptop', 50000),

(102, 'Mobile', 20000),

(103, 'Headphones', 2000),

(104, 'Keyboard', 1500),

(105, 'Mouse', 800);

.....

```
CREATE TABLE ordered_item(order_id INT,product_id INT,quantity INT,FOREIGN KEY
(order_id) REFERENCES orders(order_id),FOREIGN KEY (product_id) REFERENCES
products(product_id));
```

```
INSERT INTO ordered_item (order_id, product_id, quantity) VALUES
```

(1, 101, 1),

(1, 105, 2),

(2, 102, 1),

(3, 102, 1),

(3, 104, 1),

(4, 105, 1),

(5, 101, 1),

(6, 103, 1);

.....

```
SELECT * FROM orders;
```

```
INSERT INTO orders (order_id,customer_id,order_date,total_amount) VALUES
```

```
(35, 0035, '2024-01-13', 61300),
```

```
(36, 0036, '2024-01-13', 75600),
```

```
(37, 0037, '2024-01-14', 83400),
```

```
(38, 0038, '2024-01-14', 47900),
```

```
(39, 0039, '2024-01-14', 92500),
```

```
(40, 0040, '2024-01-15', 68800),
```

```
(41, 0041, '2024-01-15', 71300),
```

```
(42, 0042, '2024-01-15', 84700),
```

```
(43, 0043, '2024-01-16', 56200),
```

```
(44, 0044, '2024-01-16', 69400),
```

```
(45, 0045, '2024-01-16', 78100),
```

```
(46, 0046, '2024-01-17', 90800),
```

```
(47, 0047, '2024-01-17', 63500),
```

```
(48, 0048, '2024-01-17', 72900),
```

```
(49, 0049, '2024-01-18', 85400),
```

```
(50, 0050, '2024-01-18', 49100),
```

```
(51, 0051, '2024-01-18', 96700),
```

```
(52, 0052, '2024-01-19', 70200),
```

.....

```
SELECT SUM(total_amount) AS total_revenue FROM orders;
```

```
SELECT COUNT(*) AS total_customers FROM customers;
```

```
SELECT COUNT(*) AS total_orders FROM orders;
```

```
SHOW TABLES;
```

```
SELECT p.product_name,SUM(oi.quantity) AS total_sold
```

```
FROM ordered_item oi
```

```
JOIN products p ON oi.product_id = p.product_id
```

```
GROUP BY p.product_name
```

```
ORDER BY total_sold DESC;
```

```
SELECT * FROM ordered_item;
```

```
SELECT c.name,SUM(o.total_amount) AS total_spent
```

```
FROM orders o
```

```
JOIN customerS c ON o.customer_id = c.customer_id
```

```
GROUP BY c.name
```

```
ORDER BY total_spent DESC;
```

```
SELECT order_date,SUM(total_amount) AS daily_sales
```

```
FROM orders
```

```
GROUP BY order_date;
```

```
SELECT p.product_name, SUM(p.price * oi.quantity) AS revenue
```

```
FROM ordered_item oi
```

```
JOIN products p ON oi.product_id = p.product_id
```

```
GROUP BY p.product_name
```

```
ORDER BY revenue DESC;
```

```
SELECT
```

```
    MONTH(order_date) AS month,
```

```
    SUM(total_amount) AS monthly_revenue
```

```
FROM orders
```

```
GROUP BY MONTH(order_date)
```

```
ORDER BY month;
```

```
SELECT

    customer_id,

        COUNT(order_id) AS total_orders

FROM orders

GROUP BY customer_id

ORDER BY total_orders DESC;
```

```
SELECT

    customer_id,

    CASE

        WHEN COUNT(order_id) > 1 THEN 'Repeat Customer'

        ELSE 'One-time-customer'

    END AS customer_type

FROM orders

GROUP BY customer_id;
```

```
SELECT

    AVG(total_amount) AS avg_order_value

FROM orders;
```

```
SELECT

    c.city,

    SUM(o.total_amount) AS total_revenue

FROM orders o

JOIN customers c ON o.customer_id = c.customer_id

GROUP BY c.city

ORDER BY total_revenue DESC;
```

```
SELECT

    oi1.product_id,

    oi2.product_id,

    COUNT(*) AS frequency

FROM ordered_item oi1

JOIN ordered_item oi2

ON oi1.order_id = oi2.order_id

AND oi1.product_id < oi2.product_id

GROUP BY oi1.product_id, oi2.product_id

ORDER BY frequency DESC;

USE ecommerce_db;

SELECT COUNT(*) FROM ordered_item;
```

```
SELECT order_id, order_date FROM orders
```

```
WHERE YEAR(order_date) = 2027
```

```
GROUP BY order_id;
```

```
UPDATE orders
```

```
SET order_date = DATE_ADD(order_date, INTERVAL-2 YEAR)
```

```
WHERE order_id BETWEEN 2437 AND 2500
```

```
AND YEAR(order_date) = 2027;
```

```
SELECT
```

```
    customer_id,
```

```
    SUM(total_amount) AS total_spent
```

```
FROM orders
```

```
GROUP BY customer_id
```

```
ORDER BY total_spent DESC;
```

```
SELECT
```

```
    customer_id,
```

```
    MAX(order_date) AS last_purchase, -- Recency
```

COUNT(order_id) AS frequency, -- Frequency

SUM(total_amount) AS monetary -- Monetary

FROM orders

GROUP BY customer_id;

SELECT

customer_id,

COUNT(DISTINCT MONTH(order_date)) AS active_months

FROM orders

GROUP BY customer_id;

SELECT

customer_id,

SUM(total_amount) * 100.0 /

(SELECT SUM(total_amount) FROM orders) AS contribution_percent

FROM orders

GROUP BY customer_id

ORDER BY contribution_percent DESC;

SELECT

```

customer_id,

SUM(total_amount) AS total_spent,

CASE

    WHEN SUM(total_amount) > 30000 THEN 'High Value'

    WHEN SUM(total_amount) BETWEEN 10000 AND 30000 THEN 'Medium Value'

    ELSE 'Low Value'

END AS customer_segment

FROM orders

GROUP BY customer_id;

SELECT

    customer_id,

    MAX(order_date) AS last_purchase

FROM orders

GROUP BY customer_id

HAVING MAX(order_date) < '2024-03-01';

.....

```

CHAPTER 6

SOFTWARE TESTING

TESTING

6.1 GENERAL

Software testing is an important phase in the software development life cycle (SDLC) that ensures the developed application works according to the specified requirements. The main objective of testing is to identify errors, defects, and inconsistencies in the system before deployment.

Testing helps to improve software quality, reliability, performance, and security. It verifies whether all modules of the application are functioning properly and whether the system produces accurate results.

In this project, testing is performed to validate the functionality of the developed system, database connectivity, user inputs, processing logic, and output results. Different testing techniques are applied to ensure that the system is stable, efficient, and user-friendly.

The major objectives of testing are:

- To identify and fix errors in the application.
- To verify that the system meets user requirements.
- To check the accuracy of data processing.
- To ensure proper communication between software components.
- To improve system performance and reliability.
- To provide a high-quality final product.

Testing activities include test planning, test case preparation, test execution, error identification, debugging, and final validation.

6.2 DEVELOPING METHODOLOGIES

Software development methodologies define the process used to design, develop, test, and maintain software applications. A proper methodology helps developers complete projects systematically and efficiently.

For this project, the Agile Development Methodology is used.

Agile Methodology

Agile is an iterative and incremental software development approach where the project is developed in small cycles called sprints. Each sprint includes planning, development, testing, and review.

The Agile methodology allows continuous improvement and quick changes based on requirements.

Phases of Agile Methodology:

1. Requirement Analysis

In this phase, user requirements are collected and analyzed. The project objectives, system requirements, and expected outcomes are identified.

2. System Design

The system architecture, database design, user interface design, and workflow are planned.

3. Development

The actual coding process is performed. Different modules are developed and integrated into the system.

4. Testing

Each developed module is tested to identify errors and ensure that the system works correctly.

5. Deployment

After successful testing, the application is deployed for real-time usage.

6. Maintenance

Continuous monitoring, updates, and improvements are performed after deployment.

Advantages of Agile Methodology:

- Faster development process.
- Easy modification of requirements.
- Continuous testing and feedback.
- Improved software quality.
- Better communication between developers and users.

6.3 TYPES OF TESTING

Testing is classified into different types based on the purpose and level of testing. The following testing methods are performed:

1. Unit Testing

Unit testing is performed to test individual components or modules of the application separately.

The objective is to verify whether each small unit works correctly.

Example:

- Checking login module.
- Testing registration form.
- Verifying database insertion operations.

2. Integration Testing

Integration testing checks whether multiple modules work together properly.

It focuses on communication between different components of the system.

Example:

- Connecting user interface with database.
- Checking data flow between modules.

3. System Testing

System testing is performed on the complete integrated application to verify the overall functionality.

It ensures that the system satisfies all specified requirements.

Example:

- Complete application workflow testing.
- Checking all features together.

4. Functional Testing

Functional testing verifies whether the system functions according to user requirements.

It tests the input, processing, and output of the application.

Example:

- User registration.
- Data storage.
- Report generation.

5. Performance Testing

Performance testing checks the speed, stability, and response time of the application. It ensures the system can handle expected workloads.

Example:

- Checking application response time.
- Testing multiple user requests.

6. Security Testing

Security testing ensures that the application protects user information and prevents unauthorized access.

Example:

- Password protection.
- Authentication checking.
- Access control.

7. User Acceptance Testing (UAT)

User Acceptance Testing is performed by end users to verify whether the system satisfies their needs.

The application is tested in a real-world environment before final release.

8. Database Testing

Database testing verifies the correctness, integrity, and consistency of stored data.

Example:

- Checking data insertion.
- Validating database records.
- Testing data retrieval.

CHAPTER 7

SNAPSHOTS OF DASHBOARD

DASHBOARD



CHAPTER 8

CONCLUSION

- The E-Commerce Sales and Customer Analysis using MySQL and Power BI project successfully demonstrates how effective data management and visualization tools can transform raw transactional data into valuable business insights.
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CHAPTER 9

FUTURE ENHANCEMENT

9.1 FUTURE ENHANCEMENTS:

- Integration of Machine Learning models for sales forecasting and customer churn prediction.
- Implementation of real-time analytics using tools like Apache Kafka.
- Migration to cloud platforms (AWS, Azure, or Google Cloud) for better scalability.
- Development of a web-based dashboard using Power BI Service for wider accessibility.

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